ARM® Cortex®-M23

32-bit Microcontroller

NuMicro® Family
M2L31 Series BSP
Revision History

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Revision 3.04.000 (Released 2026-10-14)

1. Change CMSIS from rev. 5.1.1 to rev. 6.1.0
2. Fix Doxygen warnings, comments, and typos
3. Fix USBH driver MISRA violation
4. Fix ISP VSCode Sample code Build Error
5. Fix USBH sample code warning
6. Fix ISP sample build fail.
7. Add Bus-off sequence in CAN FD sample code
8. Create ChromiumOS EC License,md and README.md to PowerDeliveryLib Folder
9. Update UTCPD Libraruies, Fixed DRP role SRC PDOs and new create Numaker_PD30_SNK Sample Code
10. [SBOM] Complete FreeRTOS identity and scan evidence
11. Merge "Update NuEclipse project file (Semihost and RMC_MultBoot) to fix compiling error"
12. Update NuEclipse project file (Semihost and RMC_MultBoot) to fix compiling error
13. Back-sync M2L31 SBOM release artifacts
14. Optimize M2L31 SBOM audit evidence
15. Refresh M2L31 root SBOMs after evidence merge
16. Update M2L31 SBOM release artifacts
17. M2L31: Move GCC HardFault handler from retarget to startup for consistent toolchain handling
18. Update root Product and TestSample SBOMs
19. Migrate SBOM evidence and add component hashes
20. Merge "Fix Cppcheck unusedStructMember warning in retarget.c"
21. Fix Cppcheck unusedStructMember warning in retarget.c
22. Correct RTC reference definition for frequency compensation
23. Merge "Fixed typos."
24. Merge "Update startup and retarget files for MISRA-C"
25. Update BSP CHM documentation with static analysis deviations
26. Update startup and retarget files for MISRA-C
27. Fixed typos.
28. Fix USB Host library MISRA C violations
29. Fix MISRA C violations
30. Update Comment
31. Update UTCPD and TouchKey Libraries
32. Update VSCode TK libraries to use the same wchar width as Keil
33. Fix VSCode Warning
34. Fix VSCode Warning
35. Update for Driver Reference Guide
36. Update for new Host HID lib
37. Merge "Fix IAR unreachable statement warning in retarget.c"
38. Fix IAR unreachable statement warning in retarget.c
39. Merge "Update TK and UTCPD Libraries for Misra-C"
40. Update TK and UTCPD Libraries for Misra-C
41. Merge "Update NK_M2L31_TK Sample Code TK_Main.c to Fix Warning Message"
42. Update NK_M2L31_TK Sample Code TK_Main.c to Fix Warning Message
43. Merge "Update sample codes for VSCode project"
44. Fix MISRA C violations
45. Update sample codes for VSCode project

46. 1. Modify GNU linker script to preserve the non-initialized low-power SRAM region.
    2. Update LPTMR sample project files to use the dedicated AC6/GCC linker scripts.
    3. Update the Touch Key sample code and tool to fix incorrect multi-function pin settings.
47. Modify the SPI driver to fix the wrong clock divder setting for I2S slave mode
48. Modify the SPI driver to fix the wrong clock divder setting for I2S mode
49. Add SBOM files
50. Udate documents
51. Merge "USB PD: Fix typo errors in file-usb_pd_policy.c"
52. USB PD: Fix typo errors in file-usb_pd_policy.c
53. Merge "Modify the command shell to provide a request PDO command and remove some redundant commands"
54. Modify the command shell to provide a request PDO command and remove some redundant commands
55. Remove non-third-party SBOM components and update product scope and licensing information
56. Update SBOM documentation and add new components
57. Merge "Add SBOM components and licensing information for Nuvoton software"
58. PD: Modify library and headers to support changing the requested PDO
59. Add SBOM components and licensing information for Nuvoton software
60. Update NuMaker_PD30_DRP's vcpkg configuration.json
61. Update NK_M2L31_TK Sample Code Project to Use Numaker Default UART Port and Remove USBC Power Delivery GCC Projects
62. Merge "Remove obsolete DFU tools and add README for usage instructions"
63. Merge "Move the HIDTransferTest tool to Tool folder and add version to the tool"
64. Remove obsolete DFU tools and add README for usage instructions
65. Move the HIDTransferTest tool to Tool folder and add version to the tool
66. Update the PD libraries and projects to eliminate errors
67. Update PD Libraries and related Sample Code
68. Remove TouchKey GCC Project Folders and Fixed VSCode Warning Message
69. New Create Sample Code for Touch Key
70. update README.md
71. update README.md
72. Revert "Update Power Delivery Sample Code"
73. Update Power Delivery Sample Code
74. Add TouchView Tool
75. Modify PDMA EnableInt function to support two more mask at the same time
76. Merge "1.Add the linker script file for operating with LPPDMA 2.Update CAN ISP VScode project setting"
77. 1.Add the linker script file for operating with LPPDMA 2.Update CAN ISP VScode project setting
78. Modify TIMER_ResetCounter() in timer.c
79. Merge "Add the linker script file for LPSPI operating with LPPDMA"
80. Add the linker script file for LPSPI operating with LPPDMA
81. Fix GITHUB ACTION GCC Build Error
82. Fix GITHUB ACTION VSCode Build Error
83. Fix bug about return value of CLK_EnablePLL().
84. Add VSCode Quick Start Guide.pdf
85. Update VSCode Project files
86. Update UTCPD DualRole Sample code IAR Project file
87. Update source code to avoid IAR warnings

88. Merge "Fix VSCode GCC build error"
89. Fix VSCode GCC build error
90. Update the vcpkg-configuration.json files of PowerDelivery Sample code
91. Add -Wl,--no-warn-rwx-segments to VSCode projects
92. Fix: RMC IAR Build Fail
93. Merge "Fix: RMC_DualBankFwUpdate GCC Build Error"
94. Fix: RMC_DualBankFwUpdate GCC Build Error
95. Update the source and header files to eliminate IAR compiler warnings
96. Remove linker file setting from VSCode Project files
97. Modify FwUpdate Sample codes for VSCode Supporting
98. Modify USBH Sample codes for VSCode Supporting
99. Modify for VSCode Supporting
100. Modify ISP-DFU/HID/MSC Sample codes for IAR & VSCode Supporting
101. Update device name from M2L31KIDAE to M2L31SG4AE
102. Update parameter in EADC temperature sensor sample code.
103. Adding NuMaker_PD30_DRP Sample Code for Battery Charger and Synchronize Original Three DRP/SRC/SNK Sample Code for Some APIs Updated
104. Merge "Update UTCPD Sample Code to Synchronizate 64 Pins MAD025/MAD026"
105. Update UTCPD Sample Code to Synchronizate 64 Pins MAD025/MAD026
106. Fix build warning & error to support GCC 14.3.1
107. Update launch.json, tasks.json, and vcpkg-configuration.json to use cortex-debug-nuvton
108. Update rmc driver
109. Remove printf from rmc driver
110. Add RMC Dual bank sample code
111. Modify Register Lock / UnLock control flow
112. 1. Revised ui2c driver - Extended multiple bytes read api to support read single byte - Merged single byte / multiple byte api implementation. 2. Fixed ISP_I2C/ISP_SPI - Disabled IRQ until ParseCmd() is completed to prevent ISP from triggering an interrupt and fetching data prematurely. 3. Fix typo.
113. revised lpi2c driver - extend multiple bytes read api to support read single byte
114. Merge "Roll back sample code SYS_SPDMode_WakeupAndReturn to fix bug."
115. Roll back sample code SYS_SPDMode_WakeupAndReturn to fix bug.
116. Update cproject file with NuMicroM23_DFP instead of NuMicro_DFP
117. Merge "Fix the bug about the second SPD power-down mode in the sample code SYS_SPDMode_WakeupVTOR."
118. Fix the bug about the second SPD power-down mode in the sample code SYS_SPDMode_WakeupVTOR.
119. Update RMC driver for Mantis issue #0016069
120. Update parameter description for LPTPWM_EnableTrigger function
121. Update for disable LPPDMA trigger
122. Update for LPPDMA trigger setting
123. Set cmake tool to version 3.31.5 to fix long path name issue
124. Update PD source code for more close to CTS requirements.
125. Add a check for the STOP bit when I2C transmission is complete.
126. Add comment and control flow for issue #10338
127. Fix the pin setting
128. Check TimerPWM clock source
129. Correc the TimerPWM pin setting
130. Update MIRC Trim sample code
131. Fix RMC_IAP build fail

132. Add the VSCode project to SYS_TrimHIRC12M/SYS_TrimHIRC48M/SYS_TrimMIRC sample codes
133. Modify the VSCode project to remove some unaccepted statements in .cproject.yml file.
134. Modify the VSCode project to fix some error or warning messages.
135. Merge "Add the VSCode project to the bsp"
136. Add the VSCode project to the bsp
137. Add Trim sample code and fix wrong comment
138. Merge "Addd comment and control flow for issue #10338"
139. Define the base address of GPIO_INT_T.
140. Addd comment and control flow for issue #10338
141. Correct the IFA Timer PWM counter define
142. Fix the Uart clock select
143. Send NAK instead of 0 to fix hsot can't sleep issue
144. Added CMSIS document

Revision 3.03.000 (Released 2026-06-02)

1. Change CMSIS from rev. 5.1.1 to rev. 6.1.0
2. Send NAK instead of 0 to fix host can't sleep issue
3. Fix the UART clock select
4. Correct the IFA Timer PWM counter define
5. Add Trim sample code
6. Add the VSCode project to the bsp
7. Update MIRC Trim sample code
8. Update TimerPWM sample code
9. Add a check for the STOP bit when I2C transmission is complete.
10. Update PD source code for more close to CTS requirements.
11. Update UTCPD Libraries and Sample Code for Certification
12. Adding NuMaker_PD30_DRP Sample Code for Battery Charger and Synchronize ...
13. Modify Timer Driver
14. Update TK Tool
15. Update PD Libraries and related Sample Code
16. Move the HIDTransferTest tool to Tool folder and add version to the tool
17. Remove obsolete DFU tools and add README for usage instructions
18. PD: Modify library and headers to support changing the requested PDO
19. Add SBOM components and licensing information for Nuvoton software
20. Modify the command shell to provide a request PDO command and remove some ...
21. USB PD: Fix typo errors in file-usb_pd_policy.c

Revision 3.02.000 (Released 2024-12-17)

1. Updated sample code for M2L31 with only 4 KB SRAM.
2. Updated setting order of scatter-gather mode for PDMA and LPPDMA driver.
3. Updated EADC driver for ADC calibration.
4. Updated USB device sample code for V_{Bus} detect.
5. Removed LPM function from USB device sample code.
6. Changed PLL frequency setting from 240 MHz to 144 MHz for USB device samples.
7. Added RMC_DummyReadCID function for RRAM write after wake-up from Power-down mode.
8. Added I^2C error handling when error flag asserted.
9. Added channel reset macro for PDMA and LPPDMA driver.

10. Minor changes for sample code.
11. Minor bug fix.

Revision 3.01.000 (Released 2024-05-16)

1. Removed Power Level 2 from SYS driver and SYS_PowerMode sample code.
2. Removed POR disable on Startup file for GCC and IAR.
3. Updated LPADC and EADC driver to enable ADC decoding with extra one cycle to improve ADC accuracy.
4. Updated I^2C driver to support 10 bits salve address mode.
5. Added 3-wire SPI interface Macro for SPI and LPSPI drivers.
6. Added CANFD CAN mode samples.
7. Minor changes for sample code.
8. Minor bug fix.

Revision 3.00.000 (Released 2024-01-15)

1. Primary release version.

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